

Judicial nested named entity recognition method with MRC framework

Hu Zhang^{a,*,#}, Jiayu Guo^{a,#}, Yujie Wang^{a,#}, Zhen Zhang^{a,#}, Hansen Zhao^b

^a School of Computer and Information Technology, Shanxi University, Taiyuan 030006, Shanxi, China

^b Department of Computation Science and Engineering, George Institute of Technology, Atlanta, USA

ARTICLE INFO

Keywords:

Named entity recognition
Machine reading comprehension
Nested named entities
Wisdom justice

ABSTRACT

Judicial named entity recognition (JNER) is a basic task of judicial intelligence and judicial service informatization. At present, the research of JNER has attracted extensive attention. However, the existing JNER methods usually can only assign a single label to a token in the input sequence, which is not applicable to nested entities where a token may be assigned two or more different labels at the same time. Therefore, this paper introduces the machine reading comprehension (MRC) framework into JNER, and proposes a judicial nested NER method based on the MRC. Firstly, we design the question template according to the characteristics of judicial nested named entities, and construct the legal text named entity dataset in MRC format. Next, we introduce the span extraction MRC model based on the pre-trained to encode the question and text, and learn the context knowledge of the entity in the question. Finally, we extract the starting and end positions of the matching span respectively through two classifiers, to get the corresponding entities. The experimental results on the information extraction dataset in “CAIL2021” show, compared with the existing baseline models, the proposed method effectively improves the recognition effect of nested entities commonly existing in the judicial field.

1. Introduction

In recent years, with the opening of judicial big data and the development of artificial intelligence technology, “wisdom justice” has gradually become a research hotspot, which can improve the efficiency of case processing for judicial workers. Nowadays, wisdom justice services are roughly divided into three stages: (1) Assist judicial officers to handle some simple, mechanical, and tedious tasks. (2) Learn artificial intelligence technology to assist in legal trials. (3) Active carry out judicial-related convenience services in courts around the world, such as intelligent judicial trials, and legal consultation (Zhong et al., 2020; Kumar Sangaiah et al., 2020). In order to promote intelligent technology-enabled and achieve a higher level of digital justice, China Judicial Big Data Research Institute, Chinese Information Processing Society of China, Tsinghua University, and other institutions have launched the “Chinese AI & Law Challenge (CAIL)” for four consecutive years since 2018. Further, the tasks of information extraction, similar case retrieval and case label prediction have been launched recently in CAIL 2021. The information extraction is a significant research task under the wisdom justice services system, which is widely concerned in industry and academia.

Named entity recognition (NER) task in information extraction is a basic task in natural language processing, which aims to recognize the required entities from unstructured text and marks them as predefined categories, such as Person, Location, Organization, etc. (Wang et al., 2022; Li et al., 2022). In the legal text, NER’s main purpose is to extract key information entities in the case, such as the criminal suspect, the involved goods, the victim, the value of goods, etc. As a key and basic work in judicial artificial intelligence, judicial named entity recognition (JNER) has important application value in judicial judgment prediction (Zhong et al., 2020), judicial knowledge graph construction, similar cases retrieval, and so on.

At present, AI has promoted the development of various fields and the research of NER has made significant progress in some general fields, such as news, medical (Zhang et al., 2022; Luo et al., 2018) and finance. However, the NER research in the judicial field is relatively few. Compared with the general field, legal texts are more complex and specialized, and often use nested named entities to describe detailed cases. Thus, JNER is more difficult. For example, the proportion of judicial nested entities in CAIL2021 dataset is about 22%. In the theft case shown in Fig. 1, the nested types of the entities include “Victim and Location”, “Victim and Stolen Goods”, “Location and Organization”, and “Stolen

* Corresponding author.

E-mail address: zhanghu@sxu.edu.cn (H. Zhang).

School of Computer and Information Technology, No.92, Wucheng Road, Xiaodian District, Taiyuan, Shanxi 030006, China



Fig. 1. Nested types of entities in theft cases.

Goods and Crime Tools”. When dealing with named entities, the existing sequence labeling methods tend to assign a single label to a token and cannot properly handle entities with nested structures, resulting in incomplete or incorrect entity type recognition. For example, when dealing with the nested entity type “People and Location” in Fig. 1, only “The flower shop operated by victim Gao Moumou” can be identified as a Location, while the Victim “Gao Moumou” is not identified. On the one hand, it leads to a decline in the accuracy of entity recognition. On the other hand, it loses the correlation between “Gao Moumou” and “operated flower shop”, which also affects the subsequent downstream tasks. If the entity’s nesting problem can be effectively solved, the accuracy of the JNER can be further improved, which will play an important role in the actual need of judicial service and downstream tasks.

MRC task aims to understand the question and the relevant article to answer questions automatically. As a general method, MRC has become a paradigm for dealing with different problems. Many basic technologies of natural language processing tasks, such as relation extraction, event extraction and textual entailment, can achieve good results by using reading comprehension methods. Inspired by these methods, we converted NER tasks into MRC tasks. Each entity type is represented by a related question query, and entities are extracted by answering these queries in a given context. For example, extracting entities type with the “victim” is formalized as extracting answer spans to the question “Who is the victim in the case?”. By answering different independent questions, the problem of entity nesting is naturally solved.

In this paper, we employ the information extraction dataset in CAIL2021 (CAIL-IE) as the experimental dataset and explore the method of NER in the judicial field. We transform the NER task into an MRC task according to a span extraction-based MRC framework. We construct the question templates by introducing the entity’s label information and the legal prior knowledge that the entity implies. Then, combining with the proposed question, we use the span extraction MRC model based on BERT (Bidirectional Encoder Representation from Transformers) (Devlin & Chang, 2019) to extract the corresponding entities from the case description context. In this way, we can extract two overlapping entities by answering different categories of questions, and verify the effectiveness of the proposed method in CAIL-IE.

The main contributions of this paper are as follows:

- (1) We explore different question construction templates for each entity type and construct a legal named entity recognition dataset based on MRC format.
- (2) We introduce the MRC framework into the NER in the judicial field, and propose a JNER method based on the MRC, which effectively solves the problem of nested entities in legal texts.

- (3) The experimental results on the CAIL-IE dataset show that the proposed method achieves significant improvements than other state-of-the-art baselines.

2. Related work

2.1. Named entity recognition (NER)

In the early research of NER, rule-based methods (Grishman & Sundheim, 1995; Black et al., 1998) play an important role. However, these approaches usually require some domain experts to make rules manually, and this tedious manual design process is not only time-consuming and labor-intensive, but also poorly transferable. Later on, machine learning based methods have been used extensively for entity recognition tasks, such as decision tree method (DT) (Quinlan, 1986), maximum entropy model (MEM) (Borthwick & Grishman, 1999), hidden markov model (HMM) (Sun et al., 2002; Zhang et al., 2003; Jindal et al., 2020), etc. The application of these methods has effectively improved the NER effect. With the wide application of deep learning methods in natural language processing, deep learning-based NER methods have gradually become mainstream. This methods realize end-to-end entity recognition through neural networks (Sharma et al., 2021) and achieve remarkable results in general domains.

Collobert et al. (2011) first proposed a multilayer neural network structures to handle NER tasks, and Word2vec is used for word embedding to capture the semantics and context of words. Word2vec is a word embedding model, proven to be effective in representing word meanings (Agrawal & Goyal, 2021). Lample et al. (2016), Ma and Hovy (2016) added long short term memory (LSTM) (Hochreiter & Schmidhuber, 1997; Rai et al., 2022) and convolutional neural networks (CNN) (Kim, 2014) modules respectively to the LSTM-CRF model, to extract the character features of words, so that the model can automatically learn character-level and word-level representations. Rei et al. (2016) proposed a combination method of word-level and character-level vectors based on the attention mechanism to capture semantic information dynamically. In addition, there are some researches on feature extraction through dilated convolution neural networks and graph convolution neural networks (GCN), so as to better model the dependence between characters and words. Strubell et al. (2017) presented the Iterated Dilated Convolutional Neural Network model (IDCNN-CRF), which speeds up the training of the model while capturing larger local features. Cetoli et al. (2017) used a set of GCNs to study the role of dependency trees in NER. In recent years, the methods of NER by introducing lexicon information have become an interesting research focus. For example, Zhang and

Yang (2018) presented a lattice-structured LSTM model, which avoids the problem of error propagation caused by inaccurate Chinese word segmentation, and explicitly utilizes word and word sequence information. Li et al. (2020) proposed a novel FLAT model for Chinese NER, which has excellent parallelization ability and improves the recognition effect by introducing lexical information and well-designed position encoding. Aiming at the complex model structure with lexicon information, Ma et al. (2020) designed a simple and effective method for incorporating word lexicons into the character representations for Chinese NER. Liu et al. (2021) proposed a lexicon-enhanced BERT for Chinese sequence labeling (LEBERT), and the incorporation of lexicon features in the middle layer of BERT is achieved by designing a lexicon adapter. There are also some methods to improve the recognition effect of Chinese named entity by fusing multi-feature in the embedding layer of the model. Li and Meng (2021) proposed a multi-feature fusion embedding method for Chinese NER by fusing semantic, glyph and phonetic features together. Wu et al. (2021) proposed a multi-metadata embedding based cross transformer (MECT) to improve the performance of Chinese NER by fuses the information of radicals, characters and words. Liu et al. (2022) extracted the features of Chinese characters from three aspects: character features, phonetic features and text features, and combined the multi-directional feature extraction of Chinese characters with BiLSTM-CRF, and proposed a new named entity recognition method (AIP).

In recent years, nested named entity recognition has gradually become a research focus of NER tasks, and methods such as graph-based networks and sequence labeling have been proposed successively. Ju et al. (2018) proposed a Layered-BiLSTM-CRF model for dynamically stacking flat NER layers to recognize nested named entities. Each flat NER layer of the model uses bidirectional LSTM to capture sequential context. The model combines the output of the LSTM layer in the current flat NER layer to construct a new representation of the detected entities. Seq2Seq model (Straková et al., 2019) regards nested NER as a sequence-to-sequence problem, consisting of tokens input and labels output sequence. The first two methods based on sequence labeling have the problem of error propagation between layers. What's more, Luo and Zhao (2021) proposed a new bipartite flat graph network model Bi-FlaG for nested named entity recognition, which contains two modules, namely, a flat NER module for outermost entities and a graph module for all inner entities.

There is also some research work specifically on the method of NER in the judicial field. Wang et al. (2020) proposed the JCWA-DLSTM to recognize the entity of legal texts, which fuses multiple information representations of characters, words and improves the efficiency of legal entity boundary recognition. Aiming at the lack of labeling data for named entities in legal texts, Zhang et al. (2020) constructed a small-scale corpus and proposed a novel data augmentation method for JNER by fusing multiple models. According to the long length of entity names and the strong relevance between entities in judicial case documents, Guo et al. (2022) proposed a NER method for legal documents, which is based on the forward maximum matching strategy and community attention mechanism.

In a word, we have made great breakthroughs in the research of NER in the general field. The model based on the deep neural network has become the main NER solution. However, the existing solutions do not fully consider the domain characteristics of Chinese judicial text, and the nested entity recognition is mostly limited to the method of sequence labeling. Therefore, it is particularly important to recognize nested entities in the judicial field.

2.2. Machine reading comprehension (MRC)

The MRC task is to ask the machine to find the appropriate answer from the document according to the question. Thus, the researchers typically formalize MRC as a supervised learning problem about the (context, question, answer) triple. In 2016, Stanford University launched the

SQuAD (Stanford Question Answering Dataset) (Rajpurkar et al., 2016). After the public of this dataset, a series of reading comprehension models based on attention mechanisms have been proposed, such as BiDAF (Bi-Directional Attention Flow) (Seo et al., 2017), R-NET (Wang et al., 2017), QANet (Yu et al., 2018), etc. With the development of pre-trained language models, MRC models based on BERT, RoBERTa (Liu et al., 2019), and Albert (Lan et al., 2019) have emerged one after another, driving the rapid development of MRC research.

In recent years, MRC frameworks have also been introduced in many NLP tasks, such as relation extraction, event extraction, text classification, semantic role labeling, textual entailment, sentiment analysis, etc. For example, Li et al. (2019) transformed the task of relation extraction into a Multi-Turn Question Answering task. Levy et al. (2017) simplified the relationship extraction task without a pre-defined relation category to a reading comprehension task. Xiaoya Li et al. (2020) proposed a unified framework for handling NER, which transforms the NER task as an MRC task. Chen et al. (2020) proposed a Chinese event argument extraction using the MRC framework, which describes argument roles as natural language questions and achieves the recognition of argument roles by answering different questions in the context.

Based on the above work, the introduction of the MRC framework can help solve the problem of multiple subtasks of information extraction and achieve good results. Thus, this paper introduces the MRC framework into the NER, and proposes a judicial nested NER method based on the MRC.

3. Method

Aiming at the problem of nested entities in judicial field, we convert the NER task into an MRC task by introducing an MRC framework. The model structure is shown in Fig. 2, and it mainly includes two parts: (1) Dataset Construction for MRC Framework: combining entity types and judicial text data characteristics, we design the questions and mark the answer positions from the given judicial context data. (2) Judicial Nested Named Entity Recognition: the question and context are input to the BERT MRC model, and the contextual knowledge of the entities in the question is learned through encoding. We extract the spans of answers from the context, and use them as the annotated entities. The specific description is as follows:

3.1. Dataset construction for MRC framework

3.1.1. MRC questions construction

In the process of building a dataset oriented to reading comprehension, the questions will contain relevant knowledge information about the entity and will have an impact on the final result, so the generation of MRC questions is the most important step. We analyze the CAIL-IE dataset, which includes the plot description of the theft case, involving 10 types of related entities, namely “criminal suspect”, “victim”, “crime tools”, “stolen goods”, “stolen currency”, “value of goods”, “theft profits”, “time”, “location”, “organization”.

Here, we design two different question construction templates. Taking the entity type “crime tools” as an example, template 1 uses the entity type as the question prompt, and the question is set to “What are the crime tools in the case?”. Meanwhile, we analyze the legal texts of the theft case and design question template 2 by combining the characteristics of judicially nested entities with the label descriptions defined by CAIL2021. Template 2 question is set to “What is the crime tool used by the defendant in the case, which includes theft tools, vehicles, murder weapons, etc.” The question query templates constructed for different entity types are shown in Table 1 in detail.

3.1.2. MRC dataset construction

Combining the constructed questions and the characteristics of the span extraction MRC model, we construct the entity type-based NER

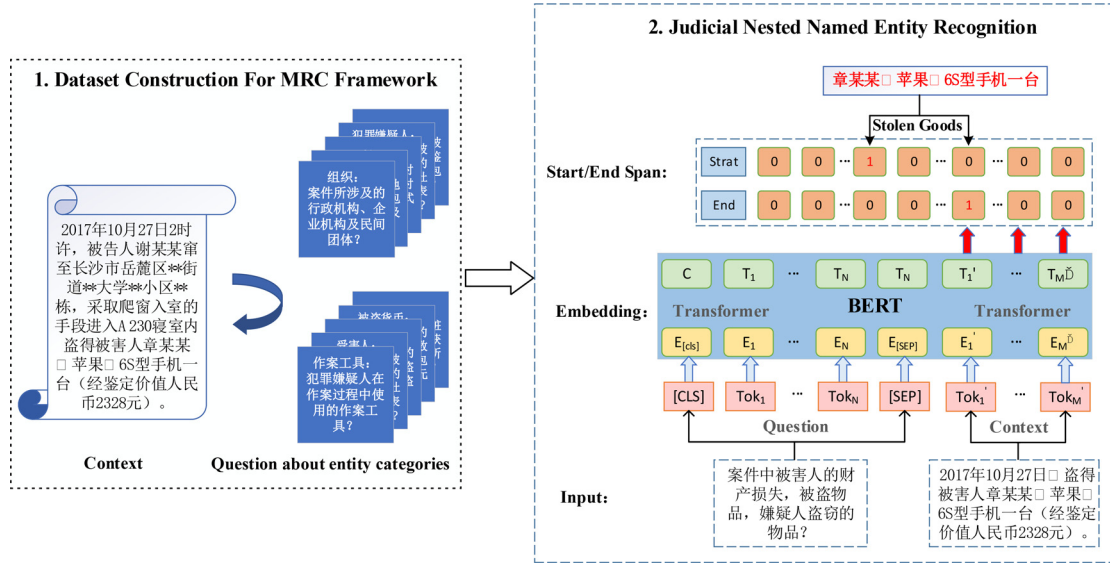


Fig. 2. The overall architecture of the proposed model.

Table 1
Conversion of different entity types to question queries.

Entities type	Template 1	Template 2
Criminal Suspect	Who is the criminal suspect in the case?	Who are the criminal suspects, defendants, and co-defendants mentioned in the case?
Victim	Who is the victim in the case?	Who are the victims and the people whose property is stolen from the case?
Crime Tools	What are the crime tools in the case?	What is the crime tool used by the defendant in the case, which includes theft tools, vehicles, murder weapons, etc?
Stolen Goods	What are the stolen goods in the case?	What are the goods stolen by the defendant and the stolen goods of the victim ?
Stolen Currency	How much money was stolen in the case?	How much money and cash are the stolen or illegally occupied property mentioned in the case?
Value of Goods	What is the value of goods in the case?	What is the appraised value and price of the stolen goods in the case?
Theft Profits	How much are the theft profits in the case?	In the case, how much profit did the defendant make from selling the illegally seized property and stolen goods?
Time	When did the case happen?	When are the period and a moment mentioned in the case?
Location	Where did the case happen?	Where are the names of streets, administrative areas, building numbers , and other locations mentioned in the case that can represent geographical location information?
Organization	What is the organization in the case?	What is the organization in the case, including administrative organizations, commercial institutions, and civil society organizations?

```

{
  "context": "1、At about 13:00 one day at the end of June,2017, the defendant Huang Mou-mou came to the entrance of Peizheng Primary School in Yuexiu District, Guangzhou City, and stole 4 batteries on Qi Moumou's electric bicycle, with a value of RMB 35 2 yuan",
  "end_position": [24],
  "entity_label": "NHCS",
  "impossible": false,
  "qas_id": "1.1",
  "query": "The code name, nicknames, social media accounts and other names indicating the identity of the suspects or defendants in the case?",
  "span_position": ["21;24"],
  "start_position": [21]
}

```

Fig. 3. Data sample in MRC format.

dataset in MRC format. Specifically, we provide a question for each entity type and mark the starting and end position of the entity in the case description, and take the entity as the answer to the question. We obtain a total of 52,470 question-answer pairs, of which 41,980 are randomly selected as the training set, 5250 are the development set, and the remaining 5240 are the test set. The data sample in MRC format is shown in Fig. 3.

3.2. Judicial nested named entity recognition

3.2.1. Encoding

For each entity type $y \in Y$, suppose the set of all entity types in the dataset is Y , the y associate with the question $q_y = \{q_1, q_2, \dots, q_m\}$. As

shown in Fig. 4, we give the input form of the input layer for several entity types such as criminal suspect, crime tools, value of goods, stolen currency, etc. That is, the formal description is that connects a case description context $X = \{x_1, x_2, \dots, x_n\}$ with the question $q_y = \{q_1, q_2, \dots, q_m\}$ and uses them together as model inputs. After encoding context and questions by the pre-trained BERT model, Then, we can obtain the embedding representation matrix T . As shown in formulas (1) - (2):

$$input = \{[CLS], q_1, q_2, \dots, q_m, [SEP], x_1, x_2, \dots, x_n, [SEP]\} \quad (1)$$

$$T = BERT(input) \quad (2)$$

Where $[CLS]$ is a classification token in BERT, $[SEP]$ is the special token to denote separation, n and m represent the length of the context, the question respectively. The $T \in \mathbb{R}^{n \times d}$, and d is the vector dimension of the last layer of BERT.

3.2.2. Entity extraction

Given a question, the span-extraction MRC model typically uses two n -class classifiers to predict the start and end positions of the answers in n characters, respectively, and only one answer span can be extracted. When converting to an NER task, the same entity type contains multiple entities and there is nesting between different entity types. Therefore, we use two binary classifiers for entity span extraction, one for predicting whether each token is a start position and the other for predicting whether each token is an end position. In a given input sequence, each token can be represented as the start and end position of an entity, and

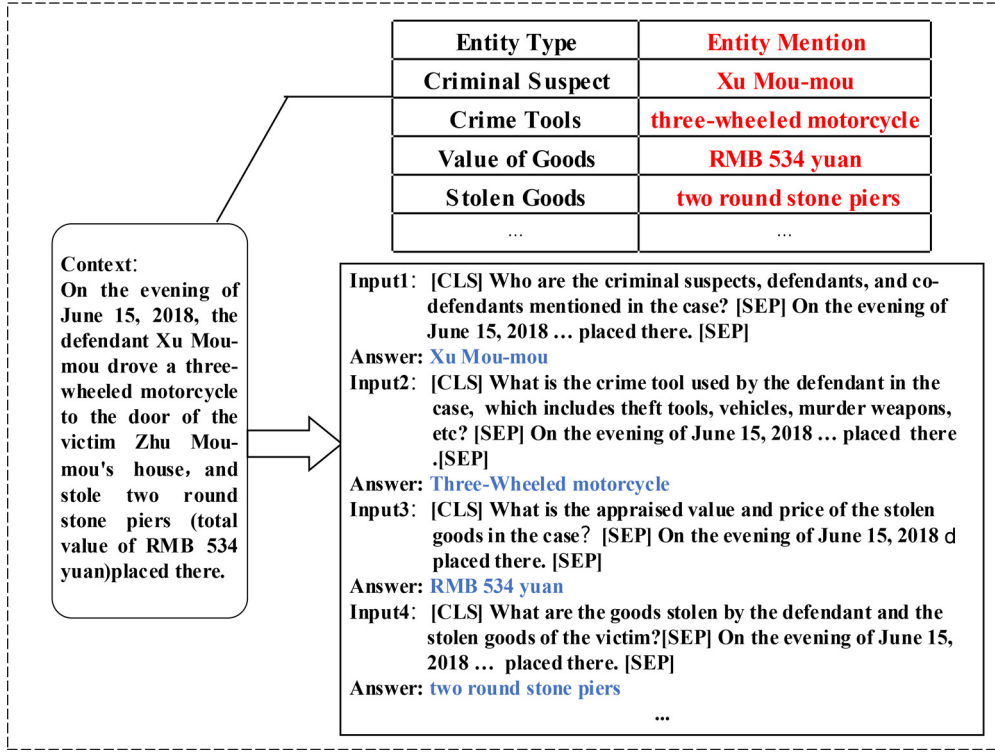


Fig. 4. Representation of the input layer.

any two tokens composed of a span can be represented as an entity, so all relevant entities in the text can be extracted according to q_y .

Taking the matrix T output by BERT as the input of the entity span extraction layer, the model predicts the probability of each token being a start position and end position. The specific calculation are shown in (3) - (4):

$$P_{start} = \text{softmax}(T \cdot W_s) \in \mathbb{R}^{n \times 2} \quad (3)$$

$$P_{end} = \text{softmax}(T \cdot W_e) \in \mathbb{R}^{n \times 2} \quad (4)$$

Where P_{start} and P_{end} present the probability of each token being elected as the start and end of the entity span, respectively. $W_s \in \mathbb{R}^{d \times 2}$ and $W_e \in \mathbb{R}^{d \times 2}$ are the weights to learn.

Multiple start and end positions are predicted, so we need to match a predicted start position with its corresponding end position. Argmax is applied to each row of P_{start} and P_{end} to get two predicted index sets of length n , which might be the starting or ending position. As shown in formulas (5) - (6):

$$\hat{I}_{start} = \{i | \arg\max(P_{start}^{(i)}) = 1, i = 1, \dots, n\} \quad (5)$$

$$\hat{I}_{end} = \{j | \arg\max(P_{end}^{(j)}) = 1, j = 1, \dots, n\} \quad (6)$$

Where indexes i, j denote the i -th and j -th row of the matrix, respectively. A binary classification layer is trained to predict the matching probability, as follows:

$$P_{i_{start}, j_{end}} = \text{sigmoid}(m \cdot \text{contact}(T_{i_{start}}, T_{j_{end}})) \quad (7)$$

Where $m \in \mathbb{R}^{1 \times 2d}$ is the weights to learns, $i_{start} \in \hat{I}_{start}$ and $j_{end} \in \hat{I}_{end}$. $T_{i_{start}}, T_{j_{end}} \in \mathbb{R}^{1 \times d}$ are the characteristic representation of the i -th and j -th characters in matrix T .

3.3. Model training

For model training, the loss is mainly divided into three parts: the entity start position prediction loss, the end position prediction loss,

and the entity start-end positions matching loss. We employ the cross-entropy loss function to calculate the loss value of the three parts, the computing formulas are shown in (8) - (10):

$$Loss_{start} = CE(P_{start}, P_{start}^*) \quad (8)$$

$$Loss_{end} = CE(P_{end}, P_{end}^*) \quad (9)$$

$$Loss_{span} = CE(P_{start, end}, Y_{start, end}) \quad (10)$$

Where P_{start}^* and P_{end}^* are the real labels of the start position and end position of the entity in the training set, respectively. $P_{start, end}^*$ present the golden labels for whether each start position should be matched with each end position. The objective training functions are defined as:

$$Loss = \alpha Loss_{start} + \beta Loss_{end} + \gamma Loss_{span} \quad (11)$$

$\alpha, \beta, \gamma \in [0, 1]$ are hyper-parameters that control the overall training objective. These three losses are jointly trained in an end-to-end manner, sharing parameters at the BERT layer.

4. Experiments

4.1. Analysis of experimental dataset

In this paper, we employ CAIL-IE as our experimental dataset, which contains 343,640 characters and 25,466 entities. The dataset is divided into training, validation, and testing sets according to a ratio of 8:1:1. It includes the plot description of the theft case, involving 10 types of related entities. Namely, Criminal Suspect, Victim, Crime Tools, Stolen Goods, Stolen Currency, Value of goods, Theft Profits, Time, Location, and Organization. The distribution of entity categories in the CAIE-IE is unbalanced, among which "Criminal Suspect" and "Stolen Goods" account for the largest proportions. The proportion of nested entities in the dataset is about 22%, and the nested entities are mostly manifested in "Person and Location", "Organizations and Location", "Location and Goods", etc. The descriptions of entity types and annotation specifications is shown in Table 2.

Table 2
Types description of entity.

Legal text	Entity label	Entity type	Eg	Proportion
At 2 o'clock on October 27, 2017, the defendant Xie Mou fled to building XX of XX Community in XX Street, Yuelu District, Changsha City. After knocking on the window with a crowbar, he stole the victim Zhang Mou's "iPhone 6s" and 1000 yuan of cash in his wallet. After winning, Xie Mou sold the stolen mobile phone for 800 RMB, and the money has been squandered. The stolen phone was valued at 4328 RMB by the Price Certification Center of Yuelu District, Changsha City.	NHVI	Victim	Zhang Mou	9%
	NCSM	Stolen Currency	1000 yuan	3%
	NCGV	Value of goods	4328 RMB	8%
	NCSP	Theft Profits	800 RMB	1%
	NATS	Crime Tools	Crowbar	2%
	NHCS	Criminal Suspect	Xie Mou	25%
	NT	Time	At 2 o'clock on October 27	11%
	NASI	Stolen Goods	Zhang Mou's iPhone 6s	23%
	NO	Organization	The Price Certification Center of Yuelu District, Changsha City	3%
	NS	Location	Building XX of XX community in XX Street, Yuelu District, Changsha City	15%

Table 3
Experimental parameter settings.

Parameter Name	Parameter Value
learning rate	1e-5
experimental environment	Tesla V100(32 G)
epoch	20
batch size	8
maximum sentence length	300
optimizer	Adam

Table 4
Experimental results of different models.

Models	P	R	F1
BiLSTM-CRF	82.33	78.61	80.43
BERT-Softmax	83.16	87.83	85.43
BERT-BiLSTM-CRF	87.79	85.32	86.54
BERT-BiLSTM-IDCNN-CRF	87.14	86.12	86.63
Ours	87.86	86.80	87.33

4.2. Experiment setting and evaluation

In this paper, we use the pre-training model (BERT-wwm) of HIT version, and the relevant parameters of the experiments are set as shown in Table 3.

The precision (P), the recall (R) and F1-values (F1) are used as evaluation metrics for NER. The computing formulas are as follows, where *correct_num*, *true_num*, *predict_num* are the number of entities extracted correctly, the number of true entities and the number of all entities extracted by the model, separately.

$$P = \frac{\text{correct_num}}{\text{predict_num}} \quad (12)$$

$$R = \frac{\text{correct_num}}{\text{true_num}} \quad (13)$$

$$F1 = \frac{2 * P * R}{P + R} \quad (14)$$

4.3. Comparison of experimental methods

In order to verify the performance of the method proposed in this paper, the method is compared with the following seven methods on the CAIL2021 information extraction dataset. These models are as follows:

- (1) BiLSTM-CRF (Ma & Hovy, 2016) models are generic NER neural models, which use the context-free vectors as input and CRF can consider the logical relationship between labels to obtain the global optimal label sequence.
- (2) BERT-BiLSTM-CRF model uses the BERT encoder to realize context-aware representation based on BiLSTM-CRF.
- (3) BERT-Softmax model in this paper is based on BERT provided by CAIL 2021 information extraction task.
- (4) BERT-BiLSTM-IDCNN-CRF model adds IDCNN with a large acceptance domain for feature extraction, so that the model can take into account both global and local features.
- (5) Layered-BiLSTM-CRF (Ju et al., 2018) recognizes nested named entities by dynamically stacking flat NER layers.
- (6) Seq2Seq model (Straková et al., 2019) regards nested NER as a sequence-to-sequence problem, consisting of tokens input and labels output sequence.

- (7) BiFlaG model (Luo & Zhao, 2021) is proposed for nested NER, which contains two modules, namely, a flat NER module for outermost entities and a graph module for all inner entities.

4.4. Experimental results and analysis

4.4.1. Entity recognition comparison

To verify the effectiveness of the proposed model on nested entity recognition, we conduct the following three groups of experiments:

- (1) For comparison, we adopt BiLSTM-CRF, BERT-softmax, BERT-BiLSTM-CRF and BERT-BiLSTM-IDCNN-CRF this four models as baselines, and the specific experimental results are shown in Table 4.

As illustrated in Table 4, compared with the BiLSTM-CRF model, the P, R, and F1 of the remaining models have been greatly improved after using the BERT pre-training model, which indicates that the pre-training model has strong text semantic representation ability. Using a pre-training model can learn rich contextual semantic information and further improves the effect of entity recognition of judicial documents. At the same time, compared with BERT-BiLSTM-CRF, the R and F1 values of BERT-BiLSTM-IDCNN-CRF are improved. It shows that although BiLSTM model can extract global context features better, it will cause local information loss, and the introduction of IDCNN can effectively improve this shortcoming. In addition, the traditional sequence labeling models may have incomplete entity recognition or wrong entity type recognition when dealing with nested entities. However, the proportion of judicial nested entities in the experimental dataset is relatively high, and our model can effectively deal with nested entities. Therefore, as shown in Table 4, the effect of our model is better than BERT-BiLSTM-CRF, BERT-softmax and BERT-BiLSTM-IDCNN-CRF, three traditional sequence labeling models.

- (2) To further verify the recognition effect of the proposed method on fine-grained entities, we conduct fine-grained recognition experiments on the BERT-BiLSTM-CRF model and our model respectively, and the results are shown in Table 5.

The experimental results show that both models have improved F1 values on most entity types. Among them, the entities of "Organization" and "Location" have increased significantly, and the F1 increased by

Table 5
Recognition results of various entities on different models.

fine-grained entities	BERT-BiLSTM-CRF			Ours		
	P	R	F1	P	R	F1
Criminal Suspect	95.35	97.55	96.43	95.91	98.12	97.00
Victim	85.31	83.89	84.59	87.31	84.88	86.08
Crime Tools	74.74	92.21	82.56	75.79	93.51	83.72
Stolen Goods	68.62	78.78	73.35	70.02	78.08	73.83
Stolen Currency	83.7	88.51	86.03	83.16	90.80	86.81
Value of goods	96.83	96.40	96.61	94.35	97.75	96.02
Theft Profits	72.55	88.10	79.57	73.80	89.28	80.81
Time	92.50	90.88	91.68	92.25	91.93	92.09
Location	76.30	83.42	79.70	79.75	83.88	81.76
Organization	76.09	92.11	83.33	78.90	93.74	85.68

2.35% and 2.06% respectively. The reason is that there are many nested “Person” and “Location” in “Organization”. For example, “(Xinzhou District, Wuhan City)” as a Location adverbial in the “(The Price Cost Supervision and Examination Bureau of Xinzhou District, Wuhan City)”, and it may be identified as a Location in the traditional models. In this case, it interferes with the correct recognition of the latter as an Organization, thereby affecting the recognition effect. Our model effectively recognizes the nested entities through question answering.

- (3) To further compare the differences between our method and other nested NER models, we conduct comparative experiments with the Layered-BiLSTM-CRF model, seq2seq model, and BiFlaG model. We compare the results of the experiments are shown in Table 6.

At present, the models for recognizing nested named entities are mainly based on sequence labeling and hyper-graph. From Table 6, we find that the model recognition based on hyper-graph is the worst. The reason is that the method based on hyper-graph needs to construct an entity graph and adjacency graph, and BiFlaG cannot recognize Chinese

Table 6
Comparison results of different nested NER models.

Model	P	R	F1
BiFlaG	63.21	71.22	66.98
Layered-BiLSTM-CRF	81.68	82.24	81.56
Seq2seq	81.10	85.62	83.30
Ours	87.86	86.80	87.33

Table 7
Comparison of experimental results for different questions.

Question Construction	P	R	F1
Template 1	87.42	86.54	86.98
Template 2	87.86	86.80	87.33

input well. Layered-BiLSTM-CRF and seq2seq, two models of sequence labeling have the problem of error propagation between layers, which leads to their poor recognition. Our model has a huge performance boost on the dataset over previous models, achieving F1 scores of 87.33%.

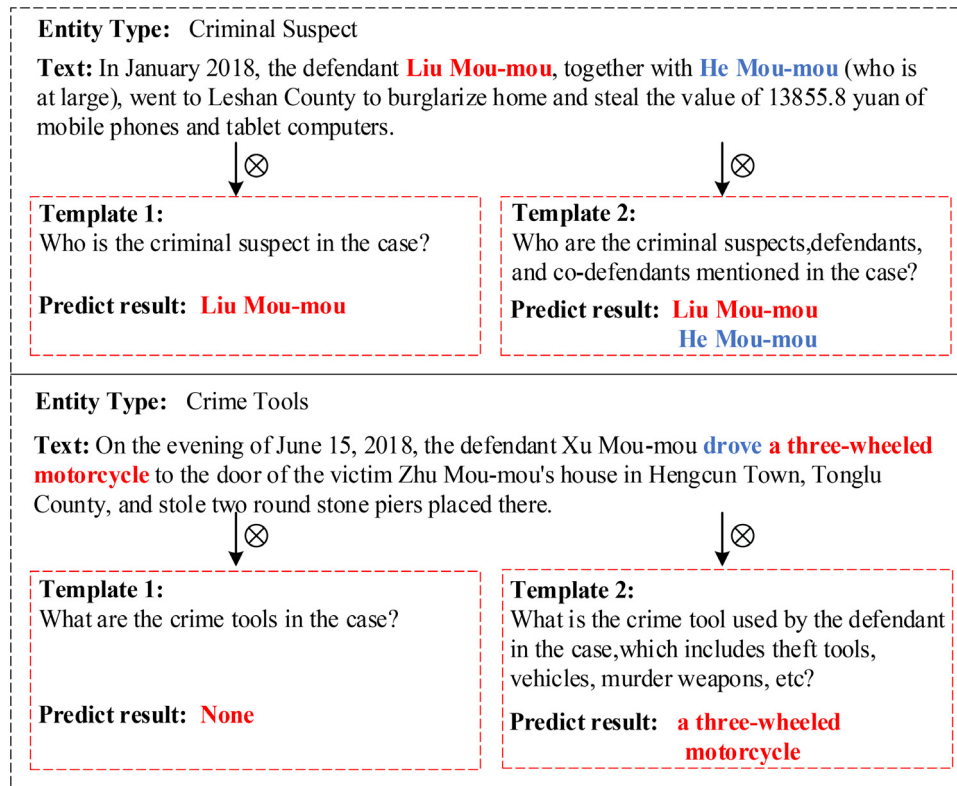


Fig. 5. Case study of entity recognition on different question templates.

Input Context	People are nested with Goods: At 2 o'clock on October 27, 2017, the defendant Xie Mou-mou fled to * * Building , * * Community, * * University, * * Street, Yuelu District, Changsha City, and entered the A230 dormitory by climbing the window to steal the victim <u>Zhang Mou-mou's</u> <u>iPhone 6S</u> (valued RMB 2328 after appraisal).	Location are nested with Organization: According to <u>The Price Cost Supervision and Examination Bureau of Xinzhou District, Wuhan City</u> , the price of the stolen office equipment is RMB 13063.
BERT-BiLSTM-CRF Model	Stolen Goods: Zhang Mou-mou's iPhone 6S Victim: None (Only the Stolen Goods were identified, but the victim was not identified)	Location: Xinzhou District, Wuhan City Organization: None (Location attributive interferes with the identification of Organization)
Ours Model	Stolen Goods: Zhang Mou-mou's iPhone 6S Victim: Zhang Mou-mou (Both the Stolen Goods and the Victims were identified)	Location: None Organization: The Price Cost Supervision and Examination Bureau of Xinzhou District, Wuhan City (Organization was identified)

Fig. 6. Nested entity case.

4.4.2. Comparison of different question templates

To further verify the influence of the constructed questions on the experimental results, we make a comparative experiment on two question templates, and the results are shown in Table 7.

From Table 7, the F1 of question template 2 is higher than that of template 1. Compared to template 1, template 2 not only asks questions about the entity labels but also considers adding the judicial semantic information and some keywords that play a role in entity recognition. For example, the important leading words "victim" and "defendant" for Victim and Criminal Suspect identification, and the keywords "value" and "price" for Value of goods, etc. At the same time, we also analyze the differences of the two question templates for the NER, using the two entity types "criminal suspect" and "crime tools" as examples, as shown in Fig. 5. The Fig. 5 and Table 7 show that the more valid information carried in the constructed questions, the richer the semantic information of the labels, and the more effective the entity extraction.

4.4.3. Case analysis

To fully illustrate the effectiveness of the proposed method in nested entity recognition, we introduce relevant examples to analyze the differences in entity recognition between BERT-BiLSTM-CRF and our method, as shown in Fig. 6.

In Fig. 6, for the same text entered, the middle is the prediction result of BERT-BiLSTM-CRF model, and the bottom is the prediction result of our model. For example, in the text on the left, the our model recognizes the nested entity "Victim" (Zhang Mou-mou) and "Stolen Goods" (Zhang Mou-mou's "iPhone 6S") in the text "Zhang Mou-mou's iPhone 6S ", but the baseline model does not recognize the victim information, which greatly reduces the accuracy of its entity recognition. Therefore, our model solves the problem of poor recognition of nested entities by introducing a MRC framework.

5. Conclusion

Aiming at the problem that the traditional NER models in the judicial field cannot effectively deal with the nested entities, we propose a judicial nested NER model based on the MRC. Firstly, the question templates are constructed based on the given judicial documents to generate the reading comprehension dataset. Then, the reading comprehension model extracts the answer spans from the context as the entity recognition result. Experimental results demonstrate that our method can effectively improve the recognition effect of nested entities commonly ex-

isting in the judicial field. Although recognizing nested named entities with the MRC framework is effective, the study still needs further improvement. In the future, we will explore the following directions:

- (1) the questions construction method proposed in this paper is limited to manual template construction. Therefore, we will consider using supervised methods to generate question queries with rich semantic information, so as to further improve the recognition effect of nested entities.
- (2) In this paper, the model takes sentences as input, but in legal context texts is tightly linked, and some entities are incorrectly recognized due to the lack of contextual information. Therefore, it will be our next research to introduce more contextual information to the model for document-level legal named entity recognition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the reviewers for their valuable comments and suggestions. This research was supported by the National Natural Science Foundation of China (No. 61936012, No. 62176145), and the Natural Science Foundation of Shanxi Province (No: 201901D111028).

References

- Zhong, H., Xiao, C., Tu, C., et al., (2020a). How does NLP benefit legal system: a summary of legal artificial intelligence. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 5218–5230).
- Kumar Sangaiah, A., Lu, H., & Hu, Q. (2020). Cognitive science and artificial intelligence for human cognition and communication. *IEEE Consumer Electronics Magazine*, 9(1), 72–73.
- Wang, Y., Tong, H., Zhu, Z., et al., (2022). Nested named entity recognition: A survey. *ACM Transactions on Knowledge Discovery From Data*, 16(6), 1–29.
- Li, J., Sun, A., Han, J., et al., (2022). A survey on deep learning for named entity recognition. *IEEE Transactions on Knowledge and Data Engineering*, 50–70.
- Zhong, H., Wang, Y., & Liu, Z. (2020b). Iteratively questioning and answering for interpretable legal judgment prediction. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(01), 1250–1257.
- Devlin, J., & Chang, M.-W. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT* (pp. 4171–4186).

- Grishman, R., & Sundheim, B. (1995). *Design of the MUC-6 Evaluation*. In *Proceedings of the Sixth Message Understanding Conference (MUC-6): Proceedings of a Conference Held in Columbia, Maryland* November 6-8, 1995.
- Black, W. J., Rinaldi, F., & Mowatt, D. (1998). FACILE: Description of the NE system used for MUC-7. In *Proceedings of the Seventh Message Understanding Conference (MUC-7): Proceedings of a Conference Held in Fairfax, Virginia* April 29 - May 1, 1998.
- Quinlan, J. R. (1986). Induction of decision trees. *Machine Learning*, 1, 81–106.
- Borthwick, A. E., & Grishman, R. (1999). *A maximum entropy approach to named entity recognition*. USA: New York University PhD Thesis.
- Sun, J., Gao, J., Zhang, L., et al., (2002). Chinese named entity identification using class-based language model. In *Proceeding of the 19th International Conference on Computational Linguistics*: 1 (pp. 1–7). Association for Computational Linguistics.
- Zhang, H., Liu, Q., Yu, H., et al., (2003). Chinese named entity recognition using role model. *International Journal of Computational Linguistics & Chinese Language Processing*, 8(2), 29–60.
- Jindal, R., Seeja, K. R., & Jain, S. (2020). Construction of domain ontology utilizing formal concept analysis and social media analytics. *International Journal of Cognitive Computing in Engineering*, 1, 62–69 ISSN 2666-3074.
- Sharma, S., Gupta, R., Bhatia, R., Toor, A. P., & Setia, H. (2021). Predicting microbial response to anthropogenic environmental disturbances using artificial neural network and multiple linear regression. *International Journal of Cognitive Computing in Engineering*, 2, 65–70 ISSN 2666-3074.
- Collobert, R., Weston, J., Bottou, L., et al., (2011). Natural language processing (almost) from scratch. In *Journal of Machine Learning Research*, 12(1), 2493–2537.
- Agrawal, R., & Goyal, R. (2021). Developing bug severity prediction models using word2vec. *International Journal of Cognitive Computing in Engineering*, 2, 104–115 ISSN 2666-3074.
- Lample, G., Ballesteros, M., Subramanian, S., et al., (2016). Neural Architectures for named entity recognition. In *Proceedings of the NAACL-HLT* (pp. 260–270).
- Ma, X., & Hovy, E. (2016a). End-to-end sequence labeling via Bi-directional LSTM-CNNs-CRF. *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, 1, 1064–1074.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- Rai, N., Kumar, D., Kaushik, N., Raj, C., & Ali, A. (2022). Fake News Classification using transformer based enhanced LSTM and BERT. *International Journal of Cognitive Computing in Engineering*, 3, 98–105 ISSN 2666-3074.
- Kim, Y. (2014). Convolutional neural networks for sentence classification. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 1746–1751).
- Rei, M., Crichton, G., & Pyysalo, S. (2016). Attending to characters in neural sequence labeling models. In *Proceedings of COLING 2016* (pp. 309–318). Technical Papers.
- Strubell, E., Verga, P., Belanger, D., et al., (2017). Fast and accurate entity recognition with iterated dilated convolutions. In *Proceedings of EMNLP* (pp. 2670–2680).
- Cetoli, A., Bragaglia, S., O'Harney, A., et al., (2017). Graph convolutional networks for named entity recognition. In *Proceedings of the 16th International Workshop on Treebanks and Linguistic Theories* (pp. 37–45).
- Zhang, Y., & Yang, J. (2018). Chinese NER using lattice LSTM. In *Proceedings of ACL* (pp. 1554–1564).
- Li, X., Yan, H., Qiu, X., & Huang, X. (2020a). FLAT: Chinese NER using flat-lattice transformer. In *Proceedings of ACL* (pp. 6836–6842).
- Ma, R., Peng, M., Zhang, Q., et al., (2020). Simplify the usage of lexicon in chinese NER. In *Proceedings of ACL* (pp. 5951–5960).
- Liu, W., Fu, X., Zhang, Y., et al., (2021). Lexicon enhanced chinese sequence labeling using BERT adapter. In *Proceedings of ACL* (pp. 5847–5858).
- J. Li, K. Meng. (2021). MFE-NER: Multi-feature fusion embedding for chinese named entity recognition. arXiv preprint arXiv: 2109.07877.
- Wu, S., Song, X., & Feng, Z. (2021). MECT: Multi-metadata embedding based cross-transformer for Chinese named entity recognition. In *Proceedings of ACL* (pp. 1529–1539).
- Liu, B., Su, Z., & Qu, G. (2022). AIP: A named entity recognition method combining glyphs and sounds. *Transactions on Asian and Low-Resource Language Information Processing*.
- Wang, D., Wang, S., Pei, W., et al., (2020). Named entity recognition method based on JLWA-DLSTM for legal instruments. *Journal of Chinese Information Processing*, 34(10), 51–58.
- Zhang, H., Gao, H., Zhou, J., et al., (2020). Applying model fusion to augment data for entity recognition in legal documents. In *Proceedings of NLPCCC* (pp. 244–255).
- Guo, L., Wang, S., Li, Y., et al., (2022). Named entity recognition in legal instruments based on matching strategy and community attention mechanism. *Journal of Chinese Information Processing*, 36(2), 85–92.
- Rajpurkar, P., Zhang, J., Lopyrev, K., et al., (2016). SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of EMNLP* (pp. 2383–2392).
- M. Seo, A. Kembhavi, A. Farhadi, et al. (2017). Bidirectional attention flow for machine comprehension. ICLR. arXiv preprint arXiv:1611.01603.
- Wang, W., Yang, N., Wei, F., et al., (2017). Gated self-matching networks for reading comprehension and question answering. In *Proceedings of ACL* (pp. 189–198).
- Yu, A. W., Dohan, D., Luong, M.-T., et al., (2018). QANet: Combining local convolution with global self-attention for reading comprehension. In *Proceedings of the 6th International Conference on Learning Representations*.
- Y. Liu, M. Ott, N. Goyal, et al. (2019). RoBERTa: A robustly optimized BERT pretraining approach. arXiv preprint arXiv: 1907.11692.
- Z. Lan, M. Chen, S. Goodman, et al. (2019). ALBERT: A lite BERT for self-supervised learning of language representations. arXiv preprint arXiv: 1909.11942.
- Li, X., Yin, F., Sun, Z., et al., (2019). Entity-relation extraction as multi-turn question answering. In *Proceedings of ACL* (pp. 1340–1350).
- Levy, O., Seo, M., Choi, E., et al., (2017). Zero-shot relation extraction via reading comprehension. In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL)* (pp. 333–342).
- Li, X., Feng, J., Meng, Y., et al., (2020b). A Unified MRC framework for named entity recognition. In *Proceedings of ACL* (pp. 5849–5859).
- Chen, M., Wu, F., Wang, Z., et al., (2020). Chinese event argument extraction using reading comprehension framework. In *Proceedings of the 19th Chinese National Conference on Computational Linguistics* (pp. 376–389).
- Ju, M., Miwa, M., & Ananiadou, S. (2018). A neural layered model for nested named entity recognition. In *Proceedings of NAACL-HLT* (pp. 1446–1459).
- Straková, J., Straka, M., & Hajic, J. (2019). Neural architectures for nested NER through linearization. In *Proceedings of ACL* (pp. 5326–5331).
- Luo, Y., & Zhao, H. (2021). Bipartite flat-graph network for nested named entity recognition. In *Proceedings of ACL* (pp. 6408–6418).
- Ma, X., & Hovy, E. (2016b). End-to-end sequence labeling via bi-directional lstm-cnns-crf. In *Proceedings of ACL* (pp. 1064–1074).
- Zhang, C., Ding, J., Zhan, J., et al., (2022). Fuzzy intelligence learning based on bounded rationality in IoT systems: A case study in Parkinson's disease. *IEEE Transactions on Computational Social Systems*.
- Luo, L., Li, N., Li, S., et al., (2018). A neural network ensemble approach for Chinese clinical named entity recognition. In *Proceedings of the Evaluation Tasks at the China Conference on Knowledge Graph and Semantic Computing (CCKS 2018): 2018* (pp. 1–6).